SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS: **Total Marks: 20**

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.
(a) Identify key components required in a navigation bar for an e-commerce website.
(b) Design a responsive navigation bar using appropriate CSS techniques.
(c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
(d) Evaluate the usability and suggest improvements for better user experience.

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.
(a) Identify suitable CSS properties for hover effects.
(b) Suggest appropriate **CSS animation or transition** for product interaction.
(c) Design a hover effect for a product card.
(d) Evaluate performance and user experience considerations.

Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.
(a) Identify layout requirements of a dashboard (rows, columns, alignment).
(b) Compare **Flexbox and Grid** based on layout needs.
(c) Choose the most suitable layout technique and justify your choice.
(d) Evaluate the scalability and responsiveness of the chosen approach.

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.
(a) Identify key sections of a blog layout.
(b) Design a mobile-first layout using CSS.
(c) Demonstrate how the layout scales for larger screens using media queries.
(d) Evaluate accessibility and readability of the design.

Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.
(a) Identify appropriate JavaScript events for menu interaction.
(b) Design event handling logic for menu toggle functionality.
(c) Implement interaction for user actions (click/hover).
(d) Evaluate performance and suggest improvements.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts

		requirements and concepts			
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

DETAILED SOLUTIONS

Assessment Tool 2 · CO-2 · Internet Programming · Design Task

The original three-page assessment paper is reproduced first without changing its content. The solutions begin on the following page and address all five questions, including design reasoning, complete runnable examples, evaluation and illustrative browser-view screenshots.

Question 1: Responsive Navigation Bar Design (4 Marks)

(a) Key components

An e-commerce navigation bar typically includes a brand/logo, links to Home and product categories, a search field, account/sign-in access, cart or bag indicator, offers/support links, and a mobile menu control. The exact set should be prioritized according to screen width and the most common user tasks.

(b)–(c) Responsive design and hamburger menu

The solution uses Flexbox for horizontal alignment, flexible wrapping for medium widths, and a media query at 640 px. On small screens, the navigation links are hidden until the hamburger button is activated. JavaScript updates aria-expanded and the button label, and selecting a navigation link closes the menu.

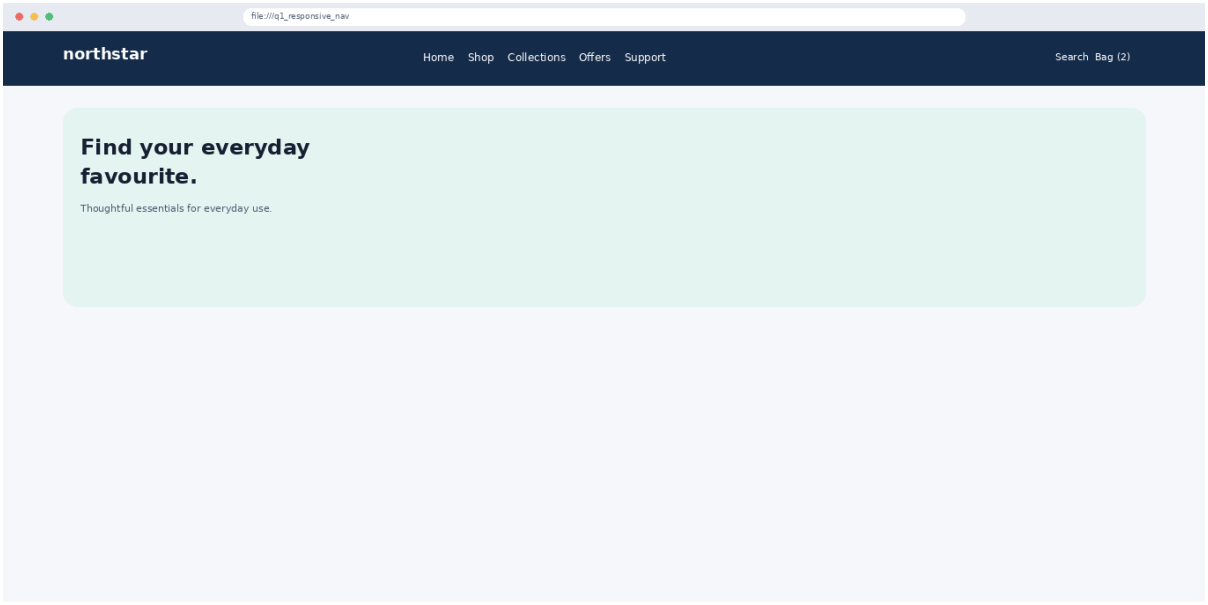
Complete code (save as q1_responsive_nav.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Northstar | Shop</title>
      <style>
        * {
        box-sizing:border-box
        }
        body {
        margin:0;
        font:16px/1.5 Arial,sans-serif;
        color:#172033;
        background:#f5f7fb
        }
        .site-header {
        background:#142b4a;
        color:#fff;
        padding:15px 5%;
        display:flex;
        align-items:center;
        gap:24px;
        justify-content:space-between
        }
        .logo {
        font-size:1.35rem;
        font-weight:800;
        white-space:nowrap
        }
        .logo span {
        color:#6dd3bd
        }
        .nav-links {
        display:flex;
        align-items:center;
        gap:22px
        }
        .nav-links a {
        color:#eaf0f8;
        text-decoration:none
        }
        .nav-links a:hover {
        color:#6dd3bd
        }
        .actions {
        display:flex;
        align-items:center;
        gap:12px
        }
        .search {
        padding:9px 12px;
        border:0;
        border-radius:7px;
        width:160px
        }
        .cart {
        color:#fff;
        text-decoration:none;
        font-weight:700
        }
        .menu-toggle {
        display:none;
        background:transparent;
        border:1px solid #ffffff66;
        color:white;
        border-radius:7px;
        padding:8px 11px;
        font-size:1.2rem;
        cursor:pointer
        }
        main {
```

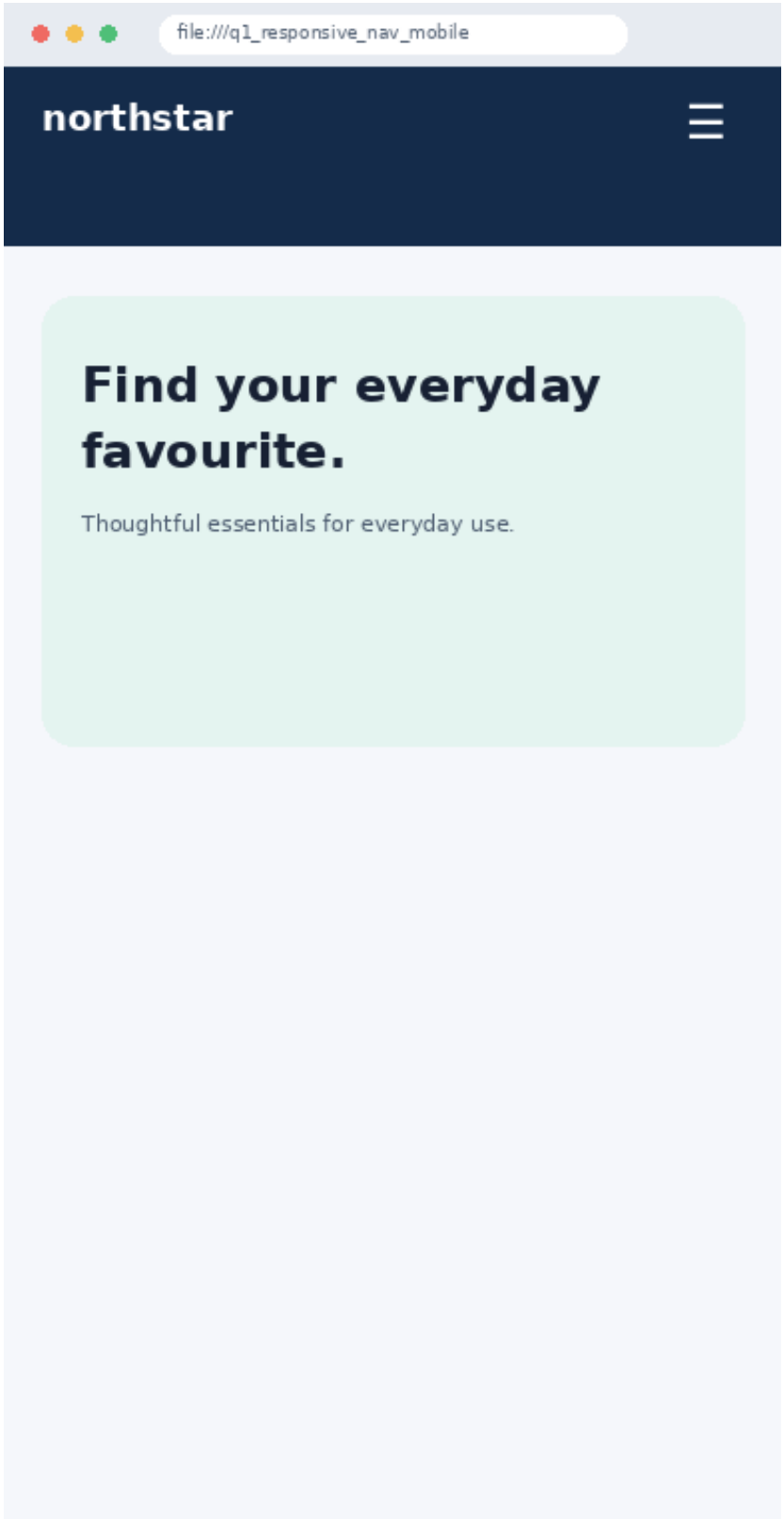
```
max-width:1100px;
margin:40px auto;
padding:0 24px
}
.hero {
background:#e4f4f0;
border-radius:18px;
padding:40px;
display:flex;
align-items:center;
justify-content:space-between;
gap:24px
}
.hero h1 {
font-size:clamp(2rem,4vw,3.2rem);
line-height:1.1;
margin:0 0 12px
}
.hero p {
color:#4d5c70;
max-width:500px
}
.button {
display:inline-block;
background:#087f73;
color:white;
text-decoration:none;
padding:11px 16px;
border-radius:8px;
font-weight:700
}
.hero-mark {
font-size:5rem;
background:#c8eae2;
border-radius:18px;
padding:24px
}
@media(max-width:900px) {
.site-header {
flex-wrap:wrap
}
.search {
width:130px
}
.nav-links {
gap:13px
}
}
@media(max-width:640px) {
.site-header {
gap:12px;
padding:14px 5%
}
.menu-toggle {
display:block;
margin-left:auto
}
.actions {
width:100%;
justify-content:space-between
}
.search {
flex:1;
min-width:0
}
.nav-links {
display:none;
flex-direction:column;
align-items:stretch;
width:100%;
gap:0
}
```

```
            .nav-links.open {
            display:flex
            }
            .nav-links a {
            padding:12px 4px;
            border-top:1px solid #ffffff22
            }
            .hero {
            padding:25px;
            flex-direction:column;
            align-items:flex-start
            }
            .hero-mark {
            font-size:3rem;
            padding:12px
            }
            .hero p {
            margin-bottom:20px
            }
            }
        </style>
    </head>
    <body>
        <header class="site-header">
            <a class="logo" href="#" style="color:white;text-decoration:none">north<span>star</span>
            </a>
            <button class="menu-toggle" id="menuToggle" aria-label="Open navigation menu" aria-expanded="false" aria-
controls="navLinks">≡</button>
            <nav class="nav-links" id="navLinks">
                <a href="#home">Home</a>
                    <a href="#shop">Shop</a>
                        <a href="#collections">Collections</a>
                            <a href="#offers">Offers</a>
                                <a href="#support">Support</a>
                                </nav>
                                <div class="actions">
                                    <input class="search" type="search" aria-label="Search products" placeholder="Search products">
                                    <a class="cart" href="#cart">Bag (2)</a>
                                    </div>
                                </header>
                                <main id="home">
                                    <section class="hero">
                                        <div>
                                            <h1>Find your everyday favourite.</h1>
                                                <p>Thoughtful essentials, carefully selected for comfort, quality and everyday use.</p>
                                                    <a class="button" href="#shop">Explore the collection</a>
                                                    </div>
                                                    <div class="hero-mark" aria-hidden="true">✦ 🛍️</div>
                                                    </section>
                                                </main>
                                                <script>
                                                    const toggle=document.getElementById('menuToggle'), nav=document.getElementById('navLinks');
                                                    toggle.addEventListener('click', ()=>{
                                                    const open=nav.classList.toggle('open');
                                                    toggle.setAttribute('aria-expanded', String(open));
                                                    toggle.setAttribute('aria-label', open?'Close navigation menu':'Open navigation menu');
                                                    toggle.textContent=open?'✕':'≡';
                                                    }
                                                    );
                                                    nav.querySelectorAll('a').forEach(link=>link.addEventListener('click', ()=>{
                                                    nav.classList.remove('open');
                                                    toggle.setAttribute('aria-expanded', 'false');
                                                    toggle.setAttribute('aria-label', 'Open navigation menu');
                                                    toggle.textContent='≡';
                                                    }
                                                    ));
                                                </script>
                                            </body>
                                        </html>
```

Screenshots



Desktop view: brand, primary navigation, search and cart are aligned in one header row.



Mobile view: the header provides a hamburger control; activate it in the supplied HTML to expand the links.

(d) Usability evaluation and improvements

The design keeps the most important shopping actions visible, provides an accessible name and expanded state for the menu button, and closes the menu after a link is selected. Further improvements include adding real destinations, a search submit action, visible keyboard focus indicators for all controls, a cart count that updates from application data, and usability tests with touch, keyboard and screen-reader users.

Question 2: Product Hover Effect (4 Marks)

(a) Suitable CSS properties

Useful properties include transform for a subtle lift or scale, box-shadow for depth, border-color for emphasis, background-color for a soft visual change, and opacity for secondary actions. These should reinforce the product rather than distract from its title, image or price.

(b) Appropriate transition

A short transition around 200–250 ms with an ease timing function is appropriate for a card lift and shadow. Avoid animating layout-affecting properties such as width, height or margin when transform can achieve the effect more smoothly.

(c) Product card implementation

The product card responds to :hover and :focus-within so keyboard users receive a similar visual cue. The design also includes a prefers-reduced-motion rule to disable transitions for users who request reduced motion.

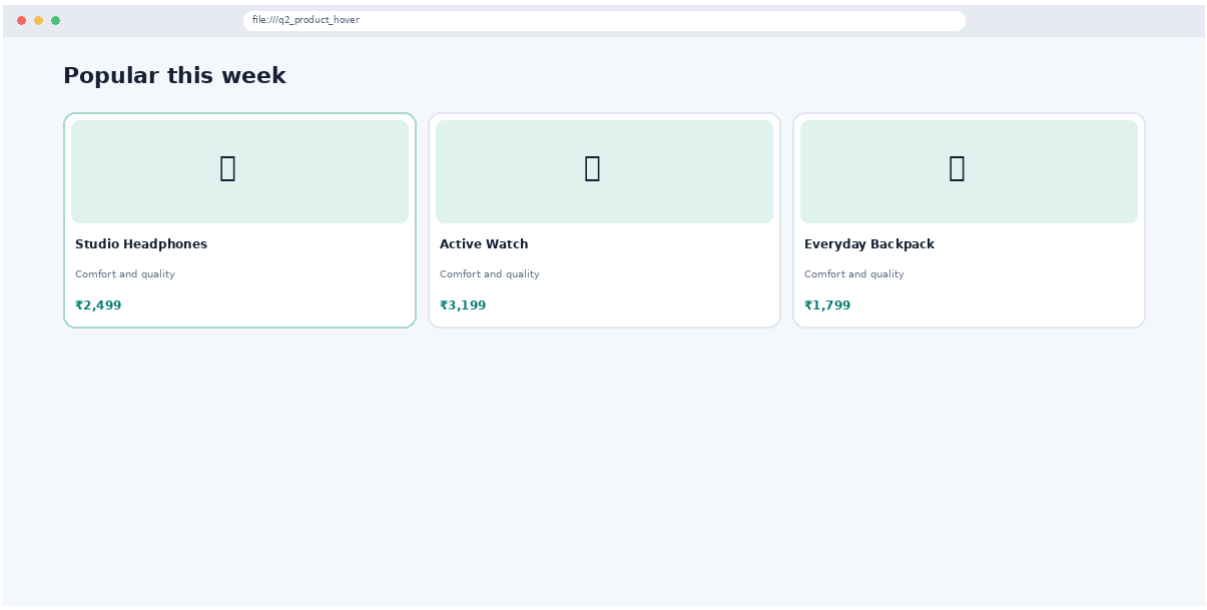
Complete code (save as q2_product_hover.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1.0">
    <title>Featured products</title>
      <style>
        * {
        box-sizing:border-box
        }
        body {
        margin:0;
        background:#f4f7fb;
        color:#172033;
        font:16px/1.5 Arial,sans-serif
        }
        main {
        max-width:1100px;
        margin:40px auto;
        padding:0 22px
        }
        h1 {
        margin-bottom:8px
        }
        .intro {
        color:#637188;
        margin-top:0
        }
        .products {
        display:grid;
        grid-template-columns:repeat(3,minmax(0,1fr));
        gap:20px;
        margin-top:25px
        }
        .card {
        background:white;
        border:1px solid #e1e7ef;
        border-radius:14px;
        overflow:hidden;
        transition:transform .22s ease,box-shadow .22s ease,border-color .22s ease
        }
        .card:hover,.card:focus-within {
        transform:translateY(-5px);
        box-shadow:0 12px 28px #18345518;
        border-color:#a7d8d0
        }
        .visual {
        height:180px;
        background:linear-gradient(135deg,#e0f2ed,#eaf0fa);
        display:grid;
        place-items:center;
        font-size:4rem;
        transition:background-color .22s ease
        }
        .card:hover .visual {
        background:#d7eee8
        }
        .details {
        padding:18px
        }
        .details h2 {
        font-size:1.1rem;
        margin:0 0 5px
        }
        .details p {
        color:#647187;
        margin:0 0 12px;
        font-size:.93rem
        }
        .price-row {
        display:flex;
        align-items:center;
```

```
        justify-content:space-between;
        gap:10px
        }
        .price {
        font-weight:800;
        color:#087f73
        }
        .view {
        background:#142b4a;
        color:white;
        border:0;
        border-radius:7px;
        padding:9px 12px;
        cursor:pointer
        }
        .view:focus-visible,.card a:focus-visible {
        outline:3px solid #f0a83b;
        outline-offset:3px
        }
        @media(max-width:760px) {
        .products {
        grid-template-columns:repeat(2,minmax(0,1fr))
        }
        }
        @media(max-width:520px) {
        .products {
        grid-template-columns:1fr
        }
        .visual {
        height:145px
        }
        }
        @media(prefers-reduced-motion:reduce) {
        .card,.visual {
        transition:none
        }
        .card:hover,.card:focus-within {
        transform:none
        }
        }
    </style>
</head>
<body>
    <main>
        <h1>Popular this week</h1>
        <p class="intro">Move your pointer over a product or navigate to its action with the keyboard.</p>
            <section class="products" aria-label="Featured products">
                <article class="card">
                    <div class="visual" aria-hidden="true">🎧</div>
                        <div class="details">
                            <h2>Studio Headphones</h2>
                            <p>Balanced sound and comfortable ear cushions.</p>
                                <div class="price-row">
                                    <span class="price">₹2,499</span>
                                    <button class="view" type="button">View item</button>
                                    </div>
                                </div>
                            </article>
                            <article class="card">
                                <div class="visual" aria-hidden="true">⌚</div>
                                    <div class="details">
                                        <h2>Active Watch</h2>
                                            <p>Simple activity tracking for busy days.</p>
                                                <div class="price-row">
                                                    <span class="price">₹3,199</span>
                                                    <button class="view" type="button">View item</button>
                                                    </div>
                                                </div>
                                            </article>
                                            <article class="card">
                                                <div class="visual" aria-hidden="true">🎒</div>
                                                    <div class="details">
```

```
                <h2>Everyday Backpack</h2>
                  <p>Lightweight storage for work and travel.</p>
                    <div class="price-row">
                      <span class="price">₹1,799</span>
                      <button class="view" type="button">View item</button>
                      </div>
                    </div>
                  </article>
                </section>
              </main>
            </body>
          </html>
```

Screenshot



Product cards with consistent spacing; the first card is highlighted to illustrate the hover/focus state.

(d) Performance and user experience

Use lightweight transitions, consistent timing and restrained shadows. Prefer transform and opacity for animated changes, avoid large continuous effects, and test on low-powered mobile devices. Hover must not be the only way to access essential actions because touchscreens have no persistent hover state. Keyboard focus styling and reduced-motion support improve accessibility.

Question 3: Layout Selection (Flexbox vs Grid) (4 Marks)

(a) Dashboard layout requirements

A dashboard usually has a row of summary metrics, a larger chart or data panel, a secondary activity panel, consistent alignment, predictable gaps and a responsive layout that can reorganize on smaller screens.

(b) Flexbox compared with Grid

Aspect	Flexbox	CSS Grid
Primary dimension	One-dimensional: row or column	Two-dimensional: rows and columns
Alignment	Excellent for distributing items along one axis	Excellent for coordinating tracks in both axes
Dashboard use	Navigation, toolbars, button groups	Metric cards and main/sidebar panels
Responsive behaviour	Items wrap or stack as needed	Tracks can resize and reorganize at breakpoints

(c) Choice and justification

CSS Grid is selected for the overall dashboard because the layout needs coordinated rows and columns: four summary cards above a main chart and a secondary activity panel. Flexbox is still useful inside individual components, such as aligning a metric label with its value or arranging activity rows.

(d) Scalability and responsiveness

The grid uses minmax(0, …) to prevent content from forcing columns wider than the available space. At 850 px, metric cards become two columns and the chart/activity area stacks into one column; at 480 px, gaps and padding are reduced. Additional cards can be added to the grid, but production dashboards should also consider loading states, long labels, chart accessibility and data density.

Complete code (save as q3_flexbox_vs_grid.html)

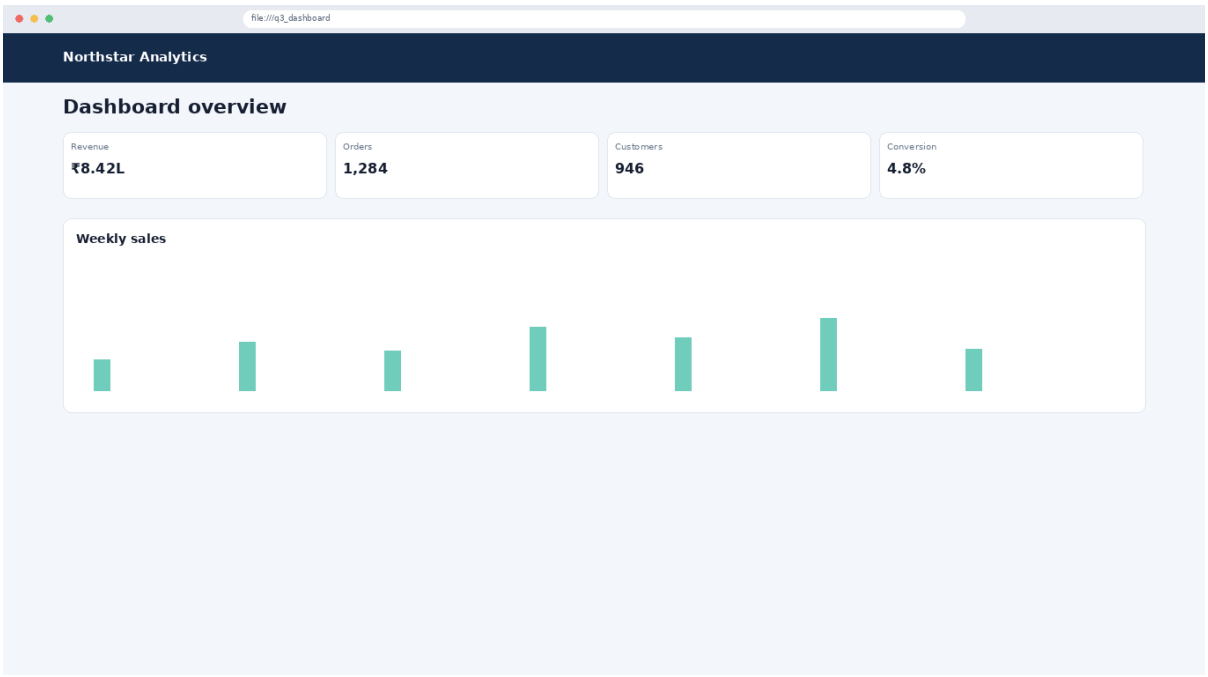
```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1.0">
    <title>Analytics dashboard</title>
      <style>
        * {
        box-sizing:border-box
        }
        body {
        margin:0;
        background:#f3f6fb;
        color:#172033;
        font:15px/1.5 Arial,sans-serif
        }
        .top {
        background:#142b4a;
        color:white;
        padding:17px 5%;
        display:flex;
        justify-content:space-between;
        align-items:center;
        gap:12px;
        flex-wrap:wrap
        }
        .top strong {
        font-size:1.2rem
        }
        .top span {
        color:#cbd8e8
        }
        .container {
        max-width:1150px;
        margin:28px auto;
        padding:0 22px
        }
        .metrics {
        display:grid;
        grid-template-columns:repeat(4,minmax(0,1fr));
        gap:15px
        }
        .metric,.panel {
        background:#fff;
        border:1px solid #e0e7f0;
        border-radius:12px;
        padding:18px;
        min-width:0
        }
        .metric .label {
        color:#68768a;
        font-size:.88rem
        }
        .metric strong {
        display:block;
        font-size:1.65rem;
        margin-top:6px
        }
        .positive {
        color:#087f73;
        font-size:.85rem
        }
        .dashboard {
        display:grid;
        grid-template-columns:minmax(0,1.7fr) minmax(250px,1fr);
        gap:18px;
        margin-top:18px;
        align-items:start
        }
        .panel h2 {
        font-size:1.1rem;
        margin:0 0 14px
```

```
}
.bars {
height:200px;
display:flex;
align-items:flex-end;
justify-content:space-around;
gap:12px;
border-bottom:1px solid #dce3ed;
padding:0 8px
}
.bar {
width:10%;
min-width:18px;
background:#70cdbb;
border-radius:6px 6px 0 0;
position:relative
}
.bar span {
position:absolute;
bottom:-26px;
left:50%;
transform:translateX(-50%);
font-size:.75rem;
color:#69768a
}
.activity {
display:flex;
align-items:center;
gap:12px;
padding:12px 0;
border-bottom:1px solid #edf0f5
}
.activity:last-child {
border:0
}
.dot {
width:10px;
height:10px;
border-radius:50%;
background:#0b8d7d;
flex-shrink:0
}
.activity small {
display:block;
color:#6b778b
}
@media(max-width:850px) {
.metrics {
grid-template-columns:repeat(2,minmax(0,1fr))
}
.dashboard {
grid-template-columns:1fr
}
}
@media(max-width:480px) {
.container {
padding:0 14px;
margin:18px auto
}
.metrics {
gap:10px
}
.metric,.panel {
padding:14px
}
.metric strong {
font-size:1.35rem
}
.bars {
gap:7px
}
}
```

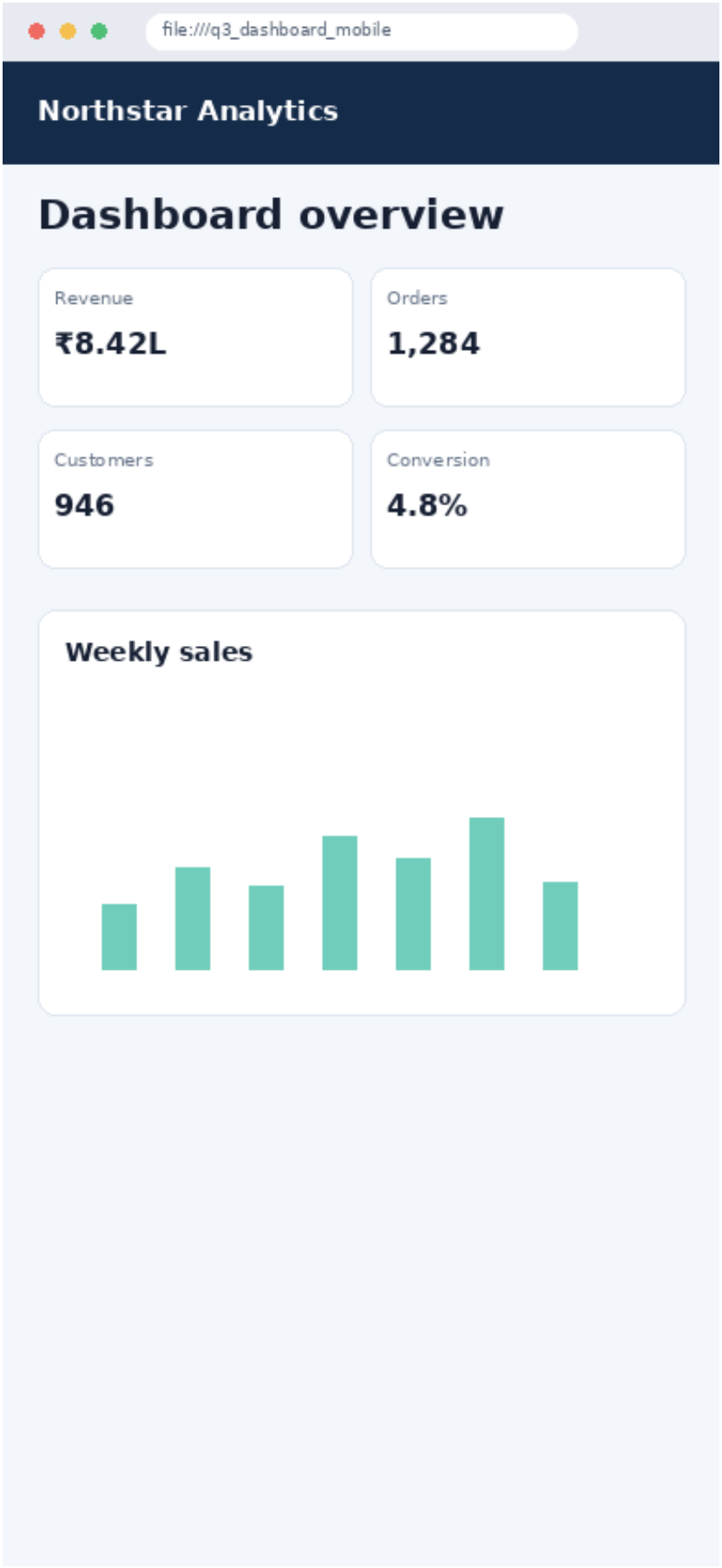
```
  </style>
</head>
<body>
  <header class="top">
    <strong>Northstar Analytics</strong>
    <span>Overview  Reports  Settings</span>
  </header>
  <main class="container">
    <h1>Dashboard overview</h1>
      <p style="color:#66748a">Performance summary · This month</p>
        <section class="metrics" aria-label="Key metrics">
          <article class="metric">
            <span class="label">Revenue</span>
            <strong>₹8.42L</strong>
            <span class="positive">↑ 12.4% this month</span>
          </article>
          <article class="metric">
            <span class="label">Orders</span>
            <strong>1,284</strong>
            <span class="positive">↑ 8.2% this month</span>
          </article>
          <article class="metric">
            <span class="label">Customers</span>
            <strong>946</strong>
            <span class="positive">↑ 5.6% this month</span>
          </article>
          <article class="metric">
            <span class="label">Conversion</span>
            <strong>4.8%</strong>
            <span class="positive">↑ 0.7% this month</span>
          </article>
        </section>
        <section class="dashboard">
          <article class="panel">
            <h2>Weekly sales</h2>
              <div class="bars" role="img" aria-label="Illustrative weekly sales chart from Monday to Sunday">
                <div class="bar" style="height:40%">
                  <span>Mon</span>
                </div>
                <div class="bar" style="height:65%">
                  <span>Tue</span>
                </div>
                <div class="bar" style="height:52%">
                  <span>Wed</span>
                </div>
                <div class="bar" style="height:82%">
                  <span>Thu</span>
                </div>
                <div class="bar" style="height:70%">
                  <span>Fri</span>
                </div>
                <div class="bar" style="height:94%">
                  <span>Sat</span>
                </div>
                <div class="bar" style="height:60%">
                  <span>Sun</span>
                </div>
              </div>
              <p style="margin-top:38px;color:#66748a;font-size:.88rem">Illustrative data for layout demonstration.</p>
              </article>
              <aside class="panel">
              <h2>Recent activity</h2>
                <div class="activity">
                  <span class="dot">
                </span>
                <div>New order received<small>Order #NS-2048 · 5 min ago</small>
                </div>
              </div>
              <div class="activity">
                <span class="dot">
              </span>
              <div>Payment completed<small>Order #NS-2046 · 18 min ago</small>
```

```
            </div>
          </div>
          <div class="activity">
            <span class="dot">
          </span>
          <div>New customer registered<small>Customer #C-912 · 1 hr ago</small>
          </div>
        </div>
      </aside>
    </section>
  </main>
 </body>
</html>
```

Screenshots



Desktop dashboard: four metrics across the top, with chart and activity panels beneath.



Mobile dashboard: metric cards use two columns and the lower panels stack vertically.

Question 4: Mobile-First Blog Layout (4 Marks)

(a) Key blog sections

A blog layout generally includes a site header and navigation, page introduction or title, main article list or featured article, metadata such as author/date/reading time, related or popular posts, optional newsletter information, and a footer.

(b) Mobile-first CSS design

The base CSS is written for narrow screens first: the article and sidebar occupy one column, spacing is compact, navigation can wrap, and typography uses relative sizing. This keeps the content readable on phones without requiring horizontal scrolling.

(c) Scaling with media queries

At 760 px and above, CSS Grid places the article and sidebar in two columns and increases padding and cover-image height. At 1000 px and above, the layout adds slightly more space and larger article headings. The max-width container prevents line lengths from becoming excessively long.

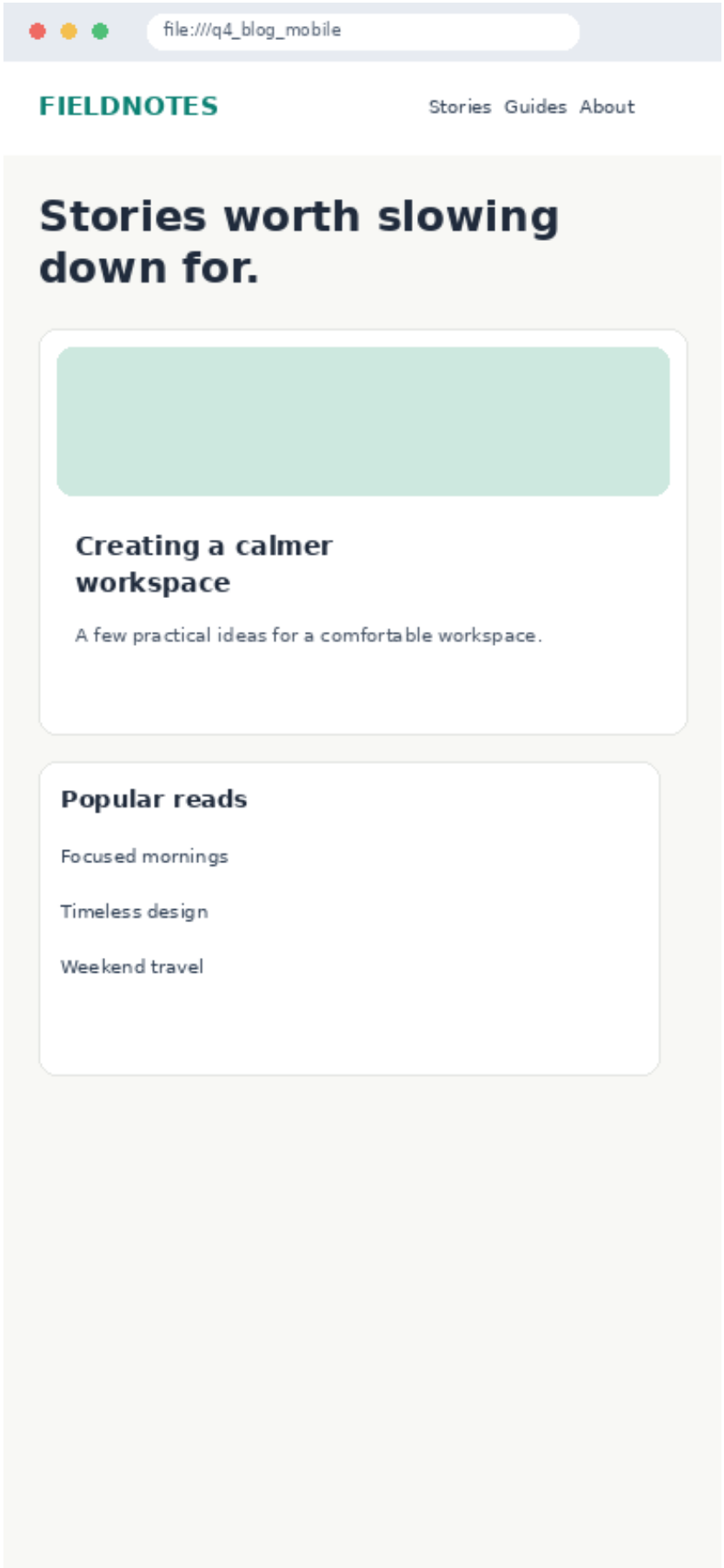
Complete code (save as q4_mobile_first_blog.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1.0">
    <title>The Field Notes Blog</title>
      <style>
        * {
        box-sizing:border-box
        }
        body {
        margin:0;
        color:#202b3c;
        background:#f8f8f5;
        font:16px/1.7 Georgia,serif
        }
        .site-header {
        padding:18px 5%;
        background:#fff;
        border-bottom:1px solid #e4e6e3;
        display:flex;
        justify-content:space-between;
        align-items:center;
        gap:12px;
        flex-wrap:wrap
        }
        .brand {
        font:800 1.2rem Arial,sans-serif;
        letter-spacing:.5px
        }
        .brand span {
        color:#0b8173
        }
        .site-header nav {
        display:flex;
        gap:16px;
        flex-wrap:wrap
        }
        .site-header a {
        color:#314156;
        text-decoration:none
        }
        .site-header a:focus-visible,.article a:focus-visible {
        outline:3px solid #0b8173;
        outline-offset:3px
        }
        .wrap {
        max-width:1050px;
        margin:26px auto;
        padding:0 18px
        }
        .intro {
        max-width:690px;
        margin:0 auto 26px;
        text-align:center
        }
        .eyebrow {
        font:700 .75rem Arial,sans-serif;
        color:#0b8173;
        letter-spacing:1.4px;
        text-transform:uppercase
        }
        .intro h1 {
        font:700 clamp(2rem,7vw,3.4rem)/1.1 Arial,sans-serif;
        margin:10px 0
        }
        .intro p {
        color:#5e6978
        }
        .layout {
        display:grid;
        grid-template-columns:minmax(0,1fr);
```

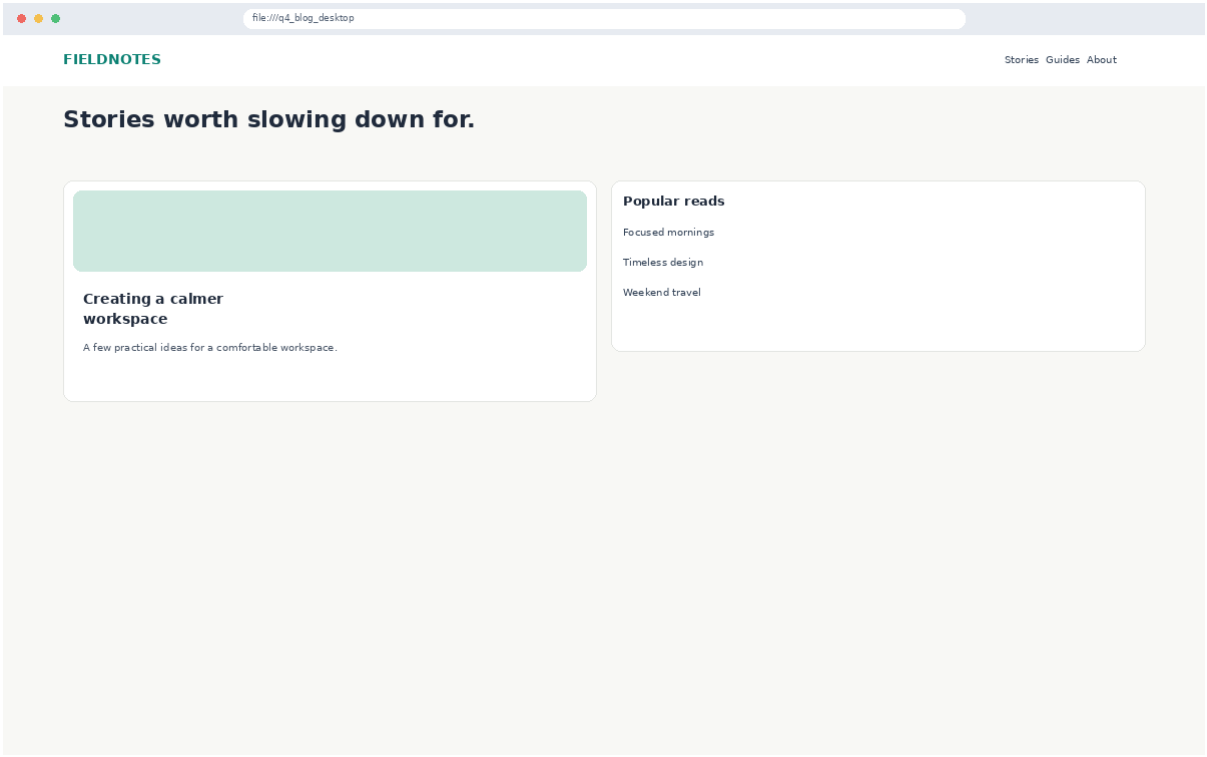
```
gap:18px
}
.article,.sidebar {
background:#fff;
border:1px solid #e4e6e3;
border-radius:12px;
padding:22px;
min-width:0
}
.cover {
height:155px;
border-radius:9px;
background:linear-gradient(135deg,#cde8df,#e9d8bd);
display:grid;
place-items:center;
font-size:3.5rem;
margin-bottom:17px
}
.article h2 {
font:700 1.45rem/1.25 Arial,sans-serif;
margin:8px 0
}
.meta {
font: .85rem Arial,sans-serif;
color:#6a7583
}
.article p {
color:#465365
}
.read {
color:#087f73;
font-weight:bold
}
.sidebar h2 {
font:700 1.1rem Arial,sans-serif;
margin-top:0
}
.sidebar ul {
padding-left:20px
}
.sidebar li {
margin:10px 0
}
.newsletter {
background:#e8f4f0;
padding:17px;
border-radius:9px;
margin-top:20px
}
.newsletter p {
margin-bottom:0;
font-size:.93rem
}
@media(min-width:760px) {
.wrap {
margin-top:38px
}
.layout {
grid-template-columns:minmax(0,2fr) minmax(230px,1fr);
gap:24px
}
.article,.sidebar {
padding:26px
}
.cover {
height:220px
}
}
@media(min-width:1000px) {
.layout {
gap:28px
}
```

```
            .article h2 {
            font-size:1.7rem
            }
            }
    </style>
  </head>
  <body>
    <header class="site-header">
      <div class="brand">FIELD<span>NOTES</span>
      </div>
      <nav aria-label="Main navigation">
        <a href="#stories">Stories</a>
          <a href="#guides">Guides</a>
            <a href="#about">About</a>
            </nav>
          </header>
          <main class="wrap">
            <section class="intro">
              <div class="eyebrow">Ideas for thoughtful living</div>
                <h1>Stories worth slowing down for.</h1>
                  <p>Practical guides, small discoveries and fresh perspectives for everyday life.</p>
                  </section>
                  <div class="layout" id="stories">
                    <article class="article">
                      <div class="cover" aria-hidden="true">🌿</div>
                        <div class="eyebrow">Design · 6 min read</div>
                          <h2>Creating a calmer, more useful workspace</h2>
                            <div class="meta">By Editorial Team · October 20, 2026</div>
                              <p>A productive workspace is not about adding more things. Start by making frequently used tools easy to reach, allowing natural light where possible, and leaving enough room for focused work.</p>
                                <p>Small improvements in lighting, storage and layout can make a workspace feel more comfortable without making it complicated.</p>
                                  <a class="read" href="#read-more">Continue reading →</a>
                                  </article>
                                  <aside class="sidebar">
                                  <h2>Popular reads</h2>
                                    <ul>
                                      <li>
                                        <a href="#habits">Five habits for a focused morning</a>
                                        </li>
                                        <li>
                                          <a href="#design">A simple guide to timeless design</a>
                                          </li>
                                          <li>
                                            <a href="#travel">Packing light for a weekend away</a>
                                            </li>
                                          </ul>
                                          <div class="newsletter">
                                            <strong>Field Notes Weekly</strong>
                                            <p>A short collection of ideas delivered every week.</p>
                                            </div>
                                          </aside>
                                        </div>
                                      </main>
                                    </body>
                                  </html>
```

Screenshots



Mobile layout: title, featured article and popular reads are arranged vertically for narrow screens.



Larger-screen layout: article content and sidebar sit side by side.

(d) Accessibility and readability

Semantic header, nav, main, article and aside elements clarify page structure. Text contrast, readable line height, responsive font sizing, descriptive link text and visible keyboard focus improve usability. Decorative imagery is marked aria-hidden. Further improvements include skip navigation, meaningful alt text for informative images, avoiding overly long line lengths, testing browser zoom, and ensuring links are distinguishable without relying only on colour.

Question 5: JavaScript Event Handling for Dynamic Menu (4 Marks)

(a) Appropriate events

The click event toggles the menu button; click events on links select a section and close the menu; mouseenter and mouseleave may provide optional hover feedback; keydown handles Escape to close the menu; and resize can reset mobile-only menu state when the viewport becomes desktop-sized. Keyboard activation of a button is supported natively by HTML.

(b)–(c) Event handling and implementation

The button toggles an open CSS class and updates aria-expanded. Menu-link clicks update the active link and status message. Escape closes the menu and returns focus to the button. Hover is treated as supplementary visual feedback rather than the only interaction method.

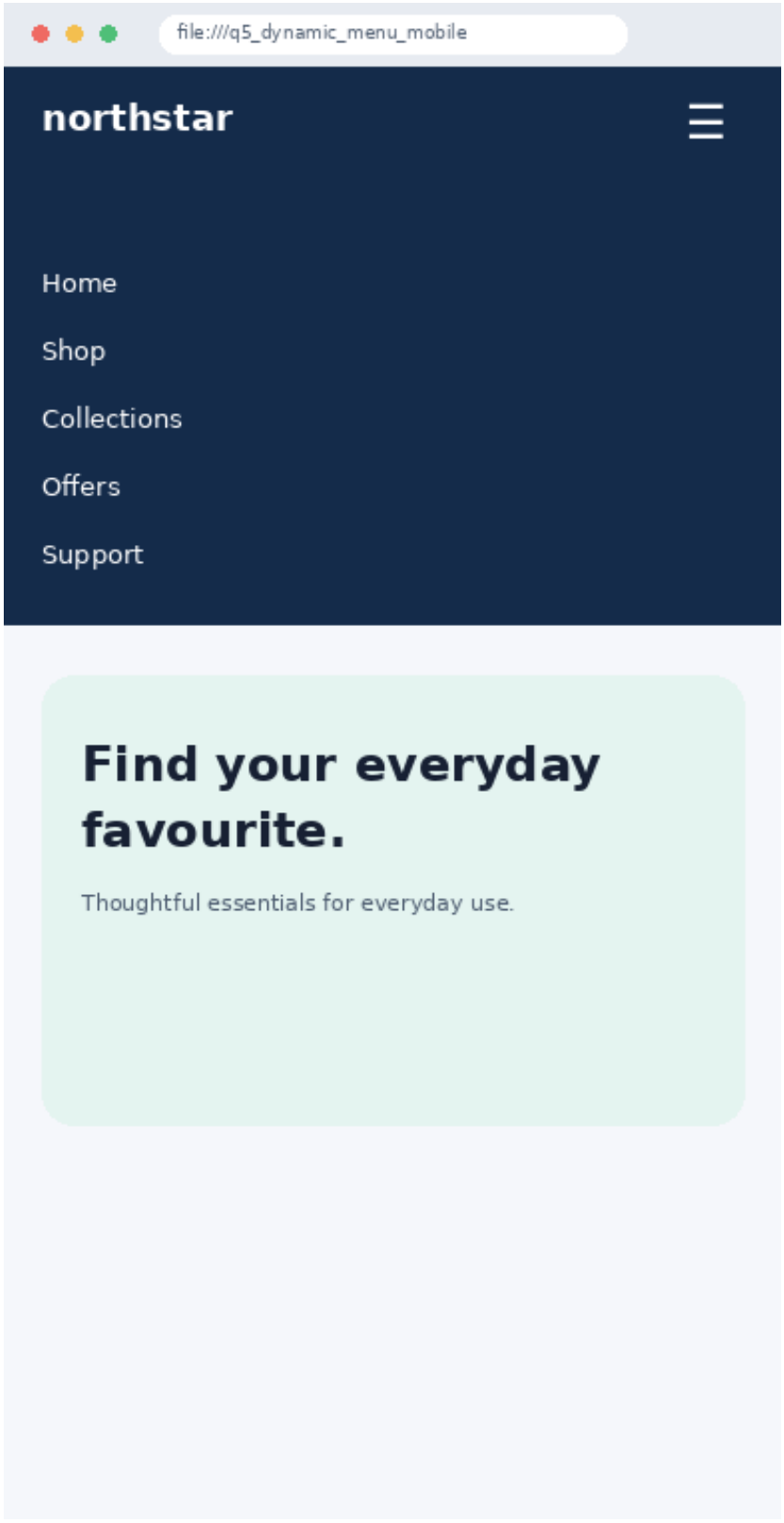
Complete code (save as q5_dynamic_menu.html)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1.0">
    <title>Interactive menu</title>
      <style>
        * {
        box-sizing:border-box
        }
        body {
        margin:0;
        background:#f4f7fb;
        color:#172033;
        font:16px/1.5 Arial,sans-serif
        }
        .header {
        background:#142b4a;
        color:white;
        padding:16px 5%;
        display:flex;
        align-items:center;
        gap:20px;
        justify-content:space-between;
        flex-wrap:wrap
        }
        .brand {
        font-weight:800;
        font-size:1.2rem
        }
        .menu-button {
        border:1px solid #ffffff66;
        background:transparent;
        color:white;
        border-radius:7px;
        padding:9px 12px;
        font-size:1rem;
        cursor:pointer
        }
        .menu {
        display:flex;
        gap:20px;
        align-items:center
        }
        .menu a {
        color:#f1f5fb;
        text-decoration:none;
        padding:7px 3px
        }
        .menu a:hover,.menu a:focus-visible {
        color:#73d7c3;
        outline-offset:4px
        }
        .menu a.active {
        border-bottom:2px solid #73d7c3
        }
        .content {
        max-width:900px;
        margin:42px auto;
        padding:0 22px
        }
        .panel {
        background:white;
        border:1px solid #e1e7ef;
        border-radius:14px;
        padding:26px
        }
        .panel h1 {
        margin-top:0
        }
        .panel p {
        color:#5d6a7d
```

```
            }
            .status {
            margin-top:16px;
            padding:12px;
            background:#e7f5f1;
            border-radius:8px;
            color:#145b51
            }
            .hint {
            color:#69768a;
            font-size:.9rem
            }
            @media(max-width:640px) {
            .header {
            gap:12px
            }
            .menu-button {
            display:block
            }
            .menu {
            display:none;
            flex-direction:column;
            align-items:stretch;
            gap:4px;
            width:100%
            }
            .menu.open {
            display:flex
            }
            .menu a {
            padding:11px 4px;
            border-top:1px solid #ffffff22
            }
            }
            @media(min-width:641px) {
            .menu-button {
            display:none
            }
            }
        </style>
    </head>
    <body>
        <header class="header">
            <div class="brand">NORTHSTAR</div>
                <button class="menu-button" id="menuButton" aria-expanded="false" aria-controls="mainMenu">≡ Menu</button>
                    <nav class="menu" id="mainMenu" aria-label="Main menu">
                        <a class="active" href="#overview">Overview</a>
                            <a href="#products">Products</a>
                                <a href="#services">Services</a>
                                    <a href="#contact">Contact</a>
                                    </nav>
                                </header>
                                <main class="content">
                                    <section class="panel">
                                        <h1>Interactive navigation</h1>
                                            <p>Use the menu button on a small screen. Select a menu item to update the active state and close the mobile menu.</p>
                                                <p class="hint">On desktop, links remain visible. On mobile, the button toggles the menu. Hover is used for visual feedback only; click and keyboard activation remain available.</p>
                                                    <div class="status" id="status" role="status" aria-live="polite">Current section: Overview</div>
                                                    </section>
                                                </main>
                                                <script>
                                                    const button=document.getElementById('menuButton'), menu=document.getElementById('mainMenu'), status=document.getElementById('status');
                                                    function setMenu(open){
                                                    menu.classList.toggle('open', open);
                                                    button.setAttribute('aria-expanded', String(open));
                                                    button.textContent=open?'✕ Close':'≡ Menu';
                                                    }
                                                    button.addEventListener('click', ()=>setMenu(!menu.classList.contains('open')));
                                                    menu.querySelectorAll('a').forEach(link=>{
```

```
                link.addEventListener('click', event=>{
                event.preventDefault();
                menu.querySelectorAll('a').forEach(a=>a.classList.remove('active'));
                link.classList.add('active');
                status.textContent='Current section: '+link.textContent;
                setMenu(false);
                }
                );
                link.addEventListener('mouseenter', ()=>{
                link.dataset.hovering='true';
                }
                );
                link.addEventListener('mouseleave', ()=>{
                delete link.dataset.hovering;
                }
                );
                }
                );
                document.addEventListener('keydown', event=>{
                if(event.key==='Escape'){
                setMenu(false);
                button.focus();
                }
                }
                );
                window.addEventListener('resize', ()=>{
                if(window.innerWidth>640)setMenu(false);
                }
                );
        </script>
    </body>
</html>
```

Screenshot



Mobile menu example with the menu expanded; in the runnable code, click the menu button to open or close it.

(d) Performance and improvements

The event handlers are attached once to known elements, avoiding repeated registration during toggles. The menu uses class changes rather than continuously changing inline styles. For production, avoid unnecessary global listeners, debounce expensive resize work if added, provide meaningful link destinations, test focus order, and ensure the menu remains usable with touch, keyboard and assistive technology.

Overall Conclusion

These five designs demonstrate responsive navigation, purposeful product interaction, dashboard layout selection, mobile-first content design and JavaScript event handling. CSS media queries and Grid/Flexbox make layouts adaptable, while semantic HTML and accessible state attributes improve usability. The examples are self-contained HTML files and can be opened directly in a browser for demonstration and testing.